EECS 836: Machine Learning
Spring 2025
Classroom: LEEP2 2300
Lecture: Monday/Friday 4:00p.m.-5:15p.m.
Office Hour: Monday 12:00p.m.-1:00p.m. or by appointment

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COURSE DESCRIPTION

Machine learning aims to understand the structure of data and fit the data into models that can be utilized by people. From automating mundane tasks to offering intelligent insights, a variety of machine learning techniques, like computer vision, natural language processing, and recommendation systems have made a massive change in every sector of the economy.

This course will explain the fundamental principles, algorithm details, and applications of multiple areas of machine learning by lectures and case studies, including classification, regression, clustering, and deep learning. After taking this course, the students should be capable of (1) describing the techniques of machine learning and making informed trade-offs on what approaches to use; (2) formulating the machine learning projects for different problems and building modeling strategies; and (3) utilizing a variety of algorithms and tools to gather, process, derive, and evaluate insights from data.

COURSE MATERIALS

- **Textbooks (optional):**

PRML	Christopher M. Bishop – Pattern Recognition and Machine Learning https://www.microsoft.com/en-us/research/people/cmbishop/prml-book/
DL	Ian Goodfellow and Yoshua Bengio and Aaron Courville – Deep Learning https://www.deeplearningbook.org/
AN	Andrew Ng – Stanford CS229 Lecture Notes https://github.com/maxim5/cs229-2018-autumn/tree/main/notes
D2L	Aston Zhang, Zack Lipton, Mu Li, Alex Smola – Dive into Deep Learning https://d2l.ai/

- **Software**: We will use python for hands-on work, tutorial will be available.

LEARNING GOALS AND OBJECTIVES

The key objectives of this course are two-fold: (1) to teach the fundamental concepts of machine learning and (2) to provide extensive hands-on experience in applying the concepts to real-world applications. The core topics to be covered in this course include data exploration, task modeling, evaluation, and visualization. This course consists of about 15 weeks of lecture. Course project will require students to write python codes to complete specific algorithms or computational tasks. Students will propose real-world machine learning problems and apply machine learning techniques to solve applications such as image recognition, web usage analysis, financial data analysis, text mining, bioinformatics, systems management, transportation analysis, and other scientific or engineering tasks. At the end of this course, students are expected to possess the fundamental skills needed to conduct their own research in machine learning or to apply machine learning techniques to interdisciplinary research fields.

GRADING POLICY

Attendance	4%
Assignments	26%
Course Project	20%
Exam I	25%
Exam II	25%

Note that the final letter grade is based on a curve. Active class participation can earn up to 5% extra credit.

COURSE WEB SITE

The CANVAS site for this course will contain lecture notes, reading materials, assignments, and notifications. It is accessible via: https://canvas.ku.edu/. You should check it frequently to remain updated. You are responsible for keeping aware of the announcements on the course web site.

ACADEMIC INTEGRITY

All work in this course is subject to the University's Academic Integrity policy. Violations of the academic integrity policy or the course collaboration policy will incur consequences according to university regulations. Penalties for academic dishonesty may lower the final grade in the course. If one student shares code or homework with another on a different team, both the donor and the recipient of the code are in violation of the academic integrity policy. On all examinations, assignments and projects, students must sign the Honor Pledge, which states, “On my honor, as a student, I have neither given nor received unauthorized aid on this academic work.” Don’t let cheating destroy your hard-earned opportunity to learn.

COURSE OUTLINE

1. Introduction
    - What is machine learning and general machine learning process
    - Real-world machine learning applications
    - Type of machine learning tasks
    - How to frame a machine learning problem

2. Data Exploration
    - Types of attributes, properties of attribute values, types of data, data quality
    - Sampling, data normalization, data cleaning, similarity measures
    - Feature selection and dimensionality reduction

3. Supervised Learning
    - Linear regression
    - Logistic regression
    - Support Vector Machines (SVMs)
    - Graphical model
    - Ensemble learning
    - Neural networks and deep learning (CNN, RNN, GNN, BERT, etc.)
    - Generative AI
    - Large Language Model (LLM)
    - Model selection and evaluation

4. Unsupervised Learning
    - K-means
    - Hierarchical clustering
    - Cluster validation
    - Gaussian mixture model and EM algorithm

5. Machine Learning Applications
    - Recommender systems
    - Natural language processing
    - Artificial intelligence in healthcare

COURSE POLICY

- **Reading Material:** Reading materials including lecture notes, software tutorials, and papers will be made available online or in class.
- **Attendance:** Regular attendance is compulsory. Students are permitted to use computers during class for note-taking and other class-related work only. Students are **not** allowed to check emails, access web sites not related to the course or work on something that is beyond the scope of this course during the class time.
- **Assignments:** You may have discussions with your class members, but you have to submit your own work. Please be sure to keep a copy of the assignment by yourself in case that there is any problem with your hand-in or you have to use it later this semester. Assignments have to be submitted **before** the beginning of the class on the specified due day. **Late submissions without appropriate reason will not be accepted.** For assignments and project reports, you are encouraged to type your work**.**

  You are **not** allowed to possess, look at, use, or in any other way derive advantage from the solutions prepared in prior years, whether these solutions are former students' work or copies of solutions that were made available by instructors.
- **Exams:** There will be **no make-up exams.** You are required to present a written proof for situations such as going to emergency room due to unexpected and serious illness. Chatting during the exam is **not** allowed. **No** collaboration between class members will be allowed during any exam. There will be **no** extra-credit project.
- **EdTech:** EdTech services, like Chegg, retain contact information of students who use their services and will release that information, which is traceable, upon request. Using these services for cheating purposes constitutes academic misconduct, which is not tolerated in the School of Engineering. It violates Article 3r, Section 6 of its Rules & Regulations, and may lead to grades of F in compromised course(s), transcript citations of academic misconduct, and suspension (even expulsion) from the University of Kansas.
- **Responsibility:** Students are responsible for reviewing the specified chapters covered by the lecture. Please note that you are responsible for the entire contents of each chapter plus any additional handouts, unless otherwise notified. Students are responsible to check Canvas frequently and keep aware of announcements to remain updated.
- **Scholastic Dishonesty Policy**: The University defines academic dishonesty as cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Scholastic dishonesty also includes, but not limited to, providing false or misleading information to receive a postponement or an extension on assignments, and submission of essentially the same written assignment for two different courses without the permission of faculty members. Interaction for the purpose of understanding a problem is not considered cheating and will be encouraged. However, the actual solution to problems must be one's own.
- **Helpful Comments:** To get full benefit out of the class you have to work regularly. Read the textbook regularly and start working on the assignments soon after they are handed out. Plan to spend at least 12 hrs a week on this class doing assignments or reading. Welcome to Machine Learning!